Divisor Channels, Prime Saturation, and Bézout Openings

An exact Erdős–Straus–Koide–period-discriminant extension

Technical research continuation from the cited canonical workbenches

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Abstract

Starting from the exact marked Erdős–Straus selector already identified with an integral terminal-matrix slice of the proposed complex-S^6 construction, we extend the arithmetic chain in four directions. First, for every coprime odd fixed-denominator shell, not only for prime denominators, we classify all oriented reciprocal solutions by divisor channels $\delta \mid n^2$ and prove a fixed-point-free complementary-channel involution. Each channel has an explicit terminal relation matrix of determinant $-\delta$, and its channel count has an exact finite Fourier expansion over $(\mathbb{Z}/R\mathbb{Z})^\times$. Second, primitive factor coordinates turn every channel into the projective incidence $nr + \delta t \equiv 0 \pmod{R}$ and into an integral Lorentz similitude of factor n^2. Third, every primitive positive Koide null ray supports infinitely many exterior prime certificates in every admissible prime progression; the single diagonal ray $[1:1]$ alone captures relative density one of the primes in every such progression. Fourth, a chosen Bézout companion opens each rank-one selector ray to an explicit nondegenerate determinant-$(-\delta)$ coefficient matrix. This matrix gives a regular ambient period torus, is related to the terminal relation matrix by a fully displayed $\mathrm{GL}_2(\mathbb{Z})$ left–right morphism, and maps by an exact orthogonal–homothetic map to the defect-one Koide hyperboloid. That exact map is proved to have no rational image point, so the existing integral defect-one counting theorem cannot be silently reinterpreted as an Erdős–Straus count. On the diagonal primitive ray, the number of exterior certificates for each individual prime is computed exactly from the factorization of $(p+1)/2$.

Source status. The imported prime selector, terminal matrix, Koide determinant map, and canonical period convention are taken from the byte-preserving current source bundle supplied with this continuation. Its embedded files verify against their recorded SHA-256 values. The original complex-threefold construction is attributed to Levent Alpöge with the disclosed Claude/Fable contribution; the separate Erdős–Straus research archive is attributed in its release to The Clankers. No theorem below uses the unverified global identification of the constructed threefold with S^6. Only the displayed arithmetic matrices and the ambient period convention are used.

1 Imported prime selector and determinant dictionaries

Let $p \equiv 1 \pmod{12}$ be prime. For

$$R_j = 4j + 3, \qquad a_j = \frac{p + R_j}{4}, \qquad 0 \le j < \frac{p-1}{4}, \tag{1}$$

the two source selector sets are

$$\mathcal{W}_0(p, j) = \{u \in \mathbb{Z}_{>0} : u \mid a_j^2,\ R_j \mid 4u + 1\}, \tag{2}$$

$$\mathcal{W}_1(p,j)=\{u\in\mathbb{Z}_{>0}:u\mid a_j^2,\ R_j\mid 4u+p\}. \tag{3}$$

They are the exterior and middle channels. With $\varepsilon\in\{0,1\}$ and $u\in\mathcal{W}_\varepsilon(p,j)$, put

$$T=\frac{4u+p^\varepsilon}{R_j},\qquad (\ell_0,\ell_1,\ell_2)=\left(\frac{T+1}{4},u-jT,1\right). \tag{4}$$

Then

$$12\ell_0-4\ell_1-3\ell_2=p^\varepsilon \tag{5}$$

and

$$\begin{pmatrix}1&-j\\0&1\end{pmatrix}\begin{pmatrix}R_j&-u\\4&-T\end{pmatrix}=\begin{pmatrix}3&-\ell_1\\4&1-4\ell_0\end{pmatrix}. \tag{6}$$

Thus the selector presentation and the terminal presentation differ by an explicit element of $\mathrm{SL}_2(\mathbb{Z})$.

The rank-one determinant model used below is

$$\iota(H,X,Y)=\begin{pmatrix}H+Y&X\\X&H-Y\end{pmatrix},\qquad \det\iota(H,X,Y)=H^2-X^2-Y^2. \tag{7}$$

The source orthogonal Koide map $\mathcal{F}:\mathbb{R}^3\to\mathbb{R}^3$ satisfies

$$Q_K(q)=\frac{2}{3}(q_1+q_2+q_3)^2-(q_1^2+q_2^2+q_3^2)=H^2-X^2-Y^2, \tag{8}$$

where

$$\begin{pmatrix}H\\X\\Y\end{pmatrix}=\begin{pmatrix}1/\sqrt{3}&1/\sqrt{3}&1/\sqrt{3}\\1/\sqrt{2}&0&-1/\sqrt{2}\\1/\sqrt{6}&-2/\sqrt{6}&1/\sqrt{6}\end{pmatrix}\begin{pmatrix}q_1\\q_2\\q_3\end{pmatrix}. \tag{9}$$

Two coefficient dictionaries are retained separately. The native symmetric coefficient matrix is

$$G(t,m,b)=\begin{pmatrix}2t&2m\\2m&b/3\end{pmatrix}, \tag{10}$$

with

$$\mathcal{L}(t,m,b)=\left(t+\frac{b}{6},2m,t-\frac{b}{6}\right)=(H,X,Y), \tag{11}$$

so

$$\det G=\frac{2}{3}(tb-6m^2). \tag{12}$$

The raw-period dictionary is

$$\mathcal{P}_K(H,X,Y)=\left(H+Y,\frac{X}{\sqrt{6}},H-Y\right)=(T_{\mathrm{raw}},M_{\mathrm{raw}},B_{\mathrm{raw}}), \tag{13}$$

and

$$T_{\mathrm{raw}}B_{\mathrm{raw}}-6M_{\mathrm{raw}}^2=H^2-X^2-Y^2. \tag{14}$$

Their exact connecting map is

$$\mathcal{D}(t,m,b)=\left(2t,\sqrt{\frac{2}{3}}\,m,\frac{b}{3}\right),\qquad \mathcal{P}_K=\mathcal{D}\circ\mathcal{L}^{-1}. \tag{15}$$

2 Complete divisor channels for a coprime odd shell

Fix integers

$$n \equiv 1 \pmod 4, \qquad R \equiv 3 \pmod 4, \qquad 0 < R < n, \qquad \gcd(n, R) = 1, \tag{16}$$

and put

$$a = \frac{n+R}{4}, \qquad N = na. \tag{17}$$

Then

$$\gcd(n, a) = \gcd(R, a) = 1. \tag{18}$$

The fixed-first-denominator equation

$$\frac{4}{n} = \frac{1}{a} + \frac{1}{y} + \frac{1}{z} \tag{19}$$

is equivalent to

$$Ryz = N(y+z) \tag{20}$$

and to

$$(Ry - N)(Rz - N) = N^2. \tag{21}$$

Definition 2.1 (oriented divisor channel)**.** For $\delta \mid n^2$, define

$$\mathscr{C}_\delta(n, R) = \left\{ (u, v) \in \mathbb{Z}^2_{>0} : uv = a^2, \quad R \mid 4u + \delta \right\}. \tag{22}$$

For (δ, u, v) in this set, define

$$d = \frac{n^2}{\delta} u, \qquad e = \delta v \tag{23}$$

and

$$y = \frac{N+d}{R}, \qquad z = \frac{N+e}{R}. \tag{24}$$

The orientation in (23) is fixed so that, when $n = p$ is prime, $\delta = 1$ is exactly the source exterior channel $d = p^2 u$.

Theorem 2.2 (complete oriented channel decomposition)**.** *The map*

$$\bigsqcup_{\delta \mid n^2} \mathscr{C}_\delta(n, R) \longrightarrow \left\{ (y, z) \in \mathbb{Z}^2_{>0} : Ryz = N(y+z) \right\} \tag{25}$$

given by (23)–(24) *is a bijection. Its inverse is*

$$d = Ry - N, \qquad e = Rz - N, \tag{26}$$

$$\kappa = \gcd(d, n^2), \qquad \delta = \frac{n^2}{\kappa}, \tag{27}$$

$$u = \frac{d}{\kappa}, \qquad v = \frac{e}{\delta}. \tag{28}$$

Under exchange of the final denominators, the channel coordinates transform as

$$(\delta, u, v) \longmapsto \left(\frac{n^2}{\delta}, v, u \right). \tag{29}$$

The involution has no fixed point. Therefore the number of unordered solutions in this fixed shell is

$$\frac{1}{2} \sum_{\delta \mid n^2} |\mathscr{C}_\delta(n, R)|. \tag{30}$$

Proof. Suppose first that (δ, u, v) satisfies (22). Then

$$de = \frac{n^2}{\delta} u \cdot \delta v = n^2 a^2 = N^2.$$

To verify the first integrality in (24), multiply its numerator by the unit δ modulo R:

$$\delta(N + d) = \delta na + n^2 u = n(\delta a + nu).$$

Because $n \equiv 4a \pmod{R}$, the final parenthesis is congruent to $a(\delta + 4u)$, hence to zero. Since δ is a unit modulo R, $R \mid N + d$. For the second numerator,

$$N + e = na + \delta v \equiv 4a^2 + \delta v = v(4u + \delta) \equiv 0 \pmod{R}.$$

Both numbers are positive, and (21) proves the reciprocal equation.

Conversely, let (y, z) satisfy (20). Division by z and by y shows $d = Ry - N > 0$ and $e = Rz - N > 0$, while (21) gives $de = n^2 a^2$. Since $\gcd(n, a) = 1$, the n-primary and a-primary parts split uniquely. In particular $\kappa = \gcd(d, n^2)$, $\delta = n^2/\kappa$, $u = d/\kappa$, and $v = e/\delta$ are positive integers with $uv = a^2$. Finally,

$$0 \equiv N + e = na + \delta v \equiv 4a^2 + \delta v = v(4u + \delta) \pmod{R}.$$

The integer v is a unit modulo R, so $R \mid 4u + \delta$. The two constructions are literal inverses.

Exchanging y, z exchanges d, e and gives (29). A fixed point would require $\delta = n$ and $u = v = a$. Its channel congruence would give

$$R \mid 4a + n = 2n + R,$$

so $R \mid 2n$. This contradicts $R > 1$, R odd, and $\gcd(R, n) = 1$. □

Corollary 2.3 (prime specialization)**.** *For $n = p$ prime, the three oriented channel orders are*

$$\delta = 1, \quad p, \quad p^2. \tag{31}$$

The first is the source exterior channel, the second is the source middle channel, and the third is the reversed exterior orientation. Hence

$$\#\{\textit{unordered solutions in the shell}\} = |\mathcal{W}_0(p, j)| + \frac{1}{2}|\mathcal{W}_1(p, j)|. \tag{32}$$

Example 2.4 (the orientation correction)**.** Take

$$n = 9, \qquad R = 7, \qquad a = 4, \qquad \delta = 3, \qquad u = 1, \qquad v = 16.$$

Then

$$d = \frac{81}{3} \cdot 1 = 27, \qquad e = 3 \cdot 16 = 48,$$

and

$$y = \frac{36 + 27}{7} = 9, \qquad z = \frac{36 + 48}{7} = 12.$$

Thus

$$\frac{4}{9} = \frac{1}{4} + \frac{1}{9} + \frac{1}{12}.$$

The inverse gives $\gcd(d, 81) = 27$ and therefore $\delta = 81/27 = 3$. Replacing this by $\gcd(d, 81)$ would reverse the channel order and would not preserve the source exterior orientation.

3 Terminal matrices and the divisor-order spectrum

For a channel state (δ, u, v), put

$$T_\delta = \frac{4u+\delta}{R}, \qquad j = \frac{R-3}{4}, \tag{33}$$

and define

$$\sigma_\delta = \begin{cases} 1, & \delta \equiv 1 \pmod 4, \\ -1, & \delta \equiv 3 \pmod 4. \end{cases} \tag{34}$$

Because $RT_\delta \equiv \delta \pmod 4$ and $R \equiv -1 \pmod 4$,

$$T_\delta + \sigma_\delta \equiv 0 \pmod 4. \tag{35}$$

Set

$$\ell_2 = \sigma_\delta, \qquad \ell_0 = \frac{T_\delta + \sigma_\delta}{4}, \qquad \ell_1 = u - jT_\delta. \tag{36}$$

Theorem 3.1 (complete terminal matrix of a divisor channel)**.** *The coordinates* (36) *are integral and satisfy*

$$12\ell_0 - 4\ell_1 - 3\ell_2 = \delta. \tag{37}$$

Moreover,

$$\begin{pmatrix} 1 & -j \\ 0 & 1 \end{pmatrix} \begin{pmatrix} R & -u \\ 4 & -T_\delta \end{pmatrix} = \underbrace{\begin{pmatrix} 3 & -\ell_1 \\ 4 & \ell_2 - 4\ell_0 \end{pmatrix}}_{\mathsf{R}_\delta} = \begin{pmatrix} 3 & -\ell_1 \\ 4 & -T_\delta \end{pmatrix}. \tag{38}$$

Consequently

$$\det \mathsf{R}_\delta = -\delta, \qquad \mathrm{SNF}(\mathsf{R}_\delta) = \mathrm{diag}(1, \delta). \tag{39}$$

Finally,

$$\ell_1 \equiv -\delta \pmod 3. \tag{40}$$

Thus the source admissibility $3 \nmid \ell_1$ holds exactly when $3 \nmid \delta$, and ℓ_2 is always odd.

Proof. Only the displayed identities are needed. Equation (35) proves $\ell_0 \in \mathbb{Z}$. Since $R = 4j+3$,

$$\begin{aligned} 12\ell_0 - 4\ell_1 - 3\ell_2 &= 3(T_\delta + \sigma_\delta) - 4(u - jT_\delta) - 3\sigma_\delta \\ &= (4j+3)T_\delta - 4u = RT_\delta - 4u = \delta. \end{aligned}$$

Left multiplication in (38) gives first row $(R - 4j, -u + jT_\delta) = (3, -\ell_1)$; its second row is unchanged. Also $\ell_2 - 4\ell_0 = -T_\delta$. The entries $3, 4$ have gcd one, so the first Smith invariant is one; the determinant gives the second. Reducing $\delta = 3T_\delta - 4\ell_1$ modulo three proves (40). □

Corollary 3.2 (complementary terminal orders)**.** *The two orientations of one unordered reciprocal solution have terminal orders*

$$\left\{\delta, \frac{n^2}{\delta}\right\}, \qquad \delta \cdot \frac{n^2}{\delta} = n^2. \tag{41}$$

The unique self-complementary order is $\delta = n$, but the oriented channel involution has no fixed state.

Remark 3.3 (scope of the terminal homology package). The canonical workbench contains a terminal chain calculation which, after its displayed geometric inputs are granted, assigns

$$(H_0, \ldots, H_6) \cong (\mathbb{Z}, \mathbb{Z}/\delta, \mathbb{Z}/\delta, \mathbb{Z}/\delta, \mathbb{Z}/\delta, 0, \mathbb{Z})$$

to a nonzero terminal value δ. In particular $\delta = 1$ is the homology-sphere package. The present theorem supplies its exact arithmetic input matrix; it does not independently reprove that every such matrix is realized by the global complex-threefold gluing.

4 Signed divisor boxes and exact Fourier counts

Factor

$$a = \prod_{q^e \| a} q^e. \tag{42}$$

Every divisor $u \mid a^2$ has a unique signed exponent vector

$$u = a \prod_{q^e \| a} q^{\beta_q}, \qquad -e \le \beta_q \le e. \tag{43}$$

Since $4a \equiv n \pmod R$, the channel condition becomes

$$\prod_{q^e \| a} q^{\beta_q} \equiv -\delta n^{-1} \pmod R. \tag{44}$$

All terms are units modulo R by (18).

Let

$$U_R = (\mathbb{Z}/R\mathbb{Z})^\times, \qquad \widehat{U}_R = \mathrm{Hom}(U_R, \mathbb{C}^\times),$$

and define

$$D_a(\chi) = \prod_{q^e \| a} \left(\sum_{\beta=-e}^{e} \chi(q)^\beta \right). \tag{45}$$

Theorem 4.1 (exact channel Fourier formula)**.** *For every $\delta \mid n^2$,*

$$C_\delta(n,R) := |\mathscr{C}_\delta(n,R)| = \frac{1}{\varphi(R)} \sum_{\chi \in \widehat{U}_R} \overline{\chi(-\delta n^{-1})}\, D_a(\chi). \tag{46}$$

If $n = \prod_{q^f \| n} q^f$, then the number of unordered reciprocal solutions in the shell is

$$\boxed{\frac{1}{2\varphi(R)} \sum_{\chi \in \widehat{U}_R} \overline{\chi(-n^{-1})}\, D_a(\chi) \prod_{q^f \| n} \left(\sum_{j=0}^{2f} \overline{\chi(q)}^{\,j} \right).} \tag{47}$$

The principal-character contribution is exactly

$$\frac{\tau(a^2)\tau(n^2)}{2\varphi(R)}. \tag{48}$$

Proof. Character orthogonality on U_R gives, for $x, c \in U_R$,

$$\mathbf{1}_{x=c} = \frac{1}{\varphi(R)} \sum_{\chi \in \widehat{U}_R} \chi(x)\overline{\chi(c)}.$$

Sum this identity over the signed box (43), with $c = -\delta n^{-1}$. The independent exponent sums factor and give (46). Summing over $\delta \mid n^2$ gives

$$\sum_{\delta \mid n^2} \overline{\chi(\delta)} = \prod_{q^f \| n} \sum_{j=0}^{2f} \overline{\chi(q)}^{\,j}.$$

The fixed-point-free involution in theorem 2.2 divides the oriented total by two. For the principal character, $D_a(1) = \tau(a^2)$ and the divisor sum is $\tau(n^2)$. □

5 Primitive rays and the exact Lorentz channel action

Lemma 5.1 (canonical primitive factor coordinates)**.** *For $u, v, a > 0$ with $uv = a^2$, there are unique positive integers r, s, t such that*

$$u = r^2 s, \qquad a = rst, \qquad v = st^2, \qquad \gcd(r, t) = 1. \tag{49}$$

Proof. Let $g = \gcd(u, a)$ and set

$$r = \frac{u}{g}, \qquad t = \frac{a}{g}, \qquad s = \frac{g}{r}.$$

Prime valuations in $u \mid a^2$ show $r \mid g$, so $s \in \mathbb{Z}_{>0}$. The displayed identities and $\gcd(r, t) = 1$ follow immediately. Conversely, $g = rs$, so the construction recovers the same triple and proves uniqueness. □

Since $n = 4rst - R$, one has the exact identity

$$nr + \delta t = t(4r^2 s + \delta) - Rr. \tag{50}$$

Because r, t are units modulo R,

$$\boxed{R \mid 4r^2 s + \delta \quad \Longleftrightarrow \quad R \mid nr + \delta t.} \tag{51}$$

Thus the channel is the inverse image of $[1 : -1]$ under the exact projective map

$$\Phi_{n,\delta} : \mathbb{P}^1(\mathbb{Z}/R\mathbb{Z}) \longrightarrow \mathbb{P}^1(\mathbb{Z}/R\mathbb{Z}), \qquad [r : t] \longmapsto [nr : \delta t], \tag{52}$$

restricted to the affine chart in which both coordinates are units.

The rank-one matrix of the ray is

$$\Gamma_{r,s,t} = s \begin{pmatrix} r \\ t \end{pmatrix} \begin{pmatrix} r & t \end{pmatrix} = \begin{pmatrix} u & a \\ a & v \end{pmatrix}. \tag{53}$$

The shifted shell matrix is

$$\Sigma_{n,\delta} = \begin{pmatrix} (n^2/\delta)u & na \\ na & \delta v \end{pmatrix}. \tag{54}$$

Let

$$M_{n,\delta} = \operatorname{diag}\left(\frac{n}{\sqrt{\delta}}, \sqrt{\delta} \right). \tag{55}$$

Then

$$\boxed{\Sigma_{n,\delta} = M_{n,\delta} \Gamma_{r,s,t} M_{n,\delta}^{\mathsf{T}}.} \tag{56}$$

Theorem 5.2 (integral Lorentz similitude)**.** *Under (7), congruence by $M_{n,\delta}$ is the integral linear map*

$$\rho_{n,\delta} = \begin{pmatrix} \dfrac{n^2/\delta + \delta}{2} & 0 & \dfrac{n^2/\delta - \delta}{2} \\ 0 & n & 0 \\ \dfrac{n^2/\delta - \delta}{2} & 0 & \dfrac{n^2/\delta + \delta}{2} \end{pmatrix}. \tag{57}$$

Its entries are integers and, with $J_L = \operatorname{diag}(1, -1, -1)$,

$$\rho_{n,\delta}^{\mathsf{T}} J_L \rho_{n,\delta} = n^2 J_L, \qquad \det \rho_{n,\delta} = n^3. \tag{58}$$

Writing

$$\xi_\delta = \log \frac{n}{\delta}, \tag{59}$$

one has

$$\rho_{n,\delta} = n \begin{pmatrix} \cosh \xi_\delta & 0 & \sinh \xi_\delta \\ 0 & 1 & 0 \\ \sinh \xi_\delta & 0 & \cosh \xi_\delta \end{pmatrix}. \tag{60}$$

Complementary channels have opposite rapidities, and the middle channel $\delta = n$ is the pure dilation nI_3.

Proof. For a diagonal matrix $\mathrm{diag}(A, B)$, congruence sends $H + Y$ to $A^2(H + Y)$, $H - Y$ to $B^2(H - Y)$, and X to ABX. Substitution of $A = n/\sqrt{\delta}$, $B = \sqrt{\delta}$ gives (57). Since n and δ are odd, both half-sums are integers. Direct multiplication gives (58). Finally,

$$n \cosh \log(n/\delta) = \frac{n^2/\delta + \delta}{2}, \qquad n \sinh \log(n/\delta) = \frac{n^2/\delta - \delta}{2}.$$

Replacing δ by n^2/δ negates the logarithm. □

The affine ray slope $\theta = t/r$ transforms under $M_{n,\delta}$ as

$$\theta \longmapsto \theta_\delta = \frac{\delta}{n}\theta. \tag{61}$$

The channel incidence (51) is exactly

$$\boxed{\theta_\delta \equiv -1 \pmod{R}.} \tag{62}$$

Thus every divisor channel is carried by its displayed Lorentz morphism to the same affine middle incidence.

6 Prime saturation on every primitive null ray

Fix a primitive positive pair (r, t) and an arithmetic progression

$$4 \mid M, \qquad \gcd(c, M) = 1, \qquad c \equiv 1 \pmod 4. \tag{63}$$

Put

$$A = 4rt, \qquad g = \gcd(A, M), \qquad M' = \frac{M}{g}. \tag{64}$$

Choose a prime R such that

$$R \equiv -c \pmod g, \qquad R \equiv 3 \pmod 4, \qquad \gcd(R, rtM) = 1. \tag{65}$$

The first congruence already implies the second because $4 \mid g$ and $c \equiv 1 \pmod 4$. Such primes R exist by Dirichlet's theorem.

Define $s_R \pmod R$ and $s_M \pmod{M'}$ by

$$4r^2 s_R \equiv -1 \pmod R, \tag{66}$$

$$\frac{A}{g} s_M \equiv \frac{c + R}{g} \pmod{M'}. \tag{67}$$

All displayed coefficients are units in their respective rings, and $\gcd(R, M') = 1$. Let s_0 be the unique class modulo RM' satisfying both. Put

$$p_0 = As_0 - R, \qquad Q = ARM' = MR\frac{A}{g}. \tag{68}$$

Theorem 6.1 (prime saturation of each primitive ray)**.** *The residue* $p_0 \pmod Q$ *is reduced. Every sufficiently large prime*

$$p \equiv p_0 \pmod Q \tag{69}$$

satisfies $p \equiv c \pmod M$ *and has an exterior Erdős–Straus certificate on the exact primitive ray* $[r : t]$*. Explicitly, write*

$$s = \frac{p+R}{4rt}, \qquad a = rst, \qquad u = r^2 s, \qquad v = st^2. \tag{70}$$

Then

$$R \mid pr + t, \qquad R \mid 4u + 1, \tag{71}$$

and the reciprocal denominators are

$$\boxed{x = rst, \qquad y = prs\,\frac{pr+t}{R}, \qquad z = st\,\frac{pr+t}{R}.} \tag{72}$$

Consequently

$$\#\{p \le X : p \equiv p_0 \pmod Q\} \sim \frac{\mathrm{Li}(X)}{\varphi(Q)}. \tag{73}$$

Within the primes $p \equiv c \pmod M$*, the constructed subfamily has relative density*

$$\frac{\varphi(M)}{\varphi(Q)} > 0. \tag{74}$$

Proof. The congruences (66)–(67) imply $As_0 - R \equiv c \pmod M$ and $4r^2 s_0 + 1 \equiv 0 \pmod R$. The modulus in (68) is exactly the change in $p = As - R$ when s is increased by RM'. To prove $\gcd(p_0, Q) = 1$, note

$$p_0 \equiv -R \pmod A, \qquad p_0 \equiv As_R \equiv -tr^{-1} \pmod R, \qquad p_0 \equiv c \pmod{M'}.$$

Each right side is a unit in the indicated modulus. Dirichlet's theorem, or the prime number theorem in arithmetic progressions, gives (73).

For such a prime, (70) is integral and gives $4a = p + R$ and $uv = a^2$. The first congruence in (71) follows from

$$pr + t = t(4r^2 s + 1) - Rr.$$

The second is equivalent because t is a unit modulo R. Formula (72) is the source factor-coordinate reconstruction; direct substitution also gives $4xyz = p(xy + xz + yz)$. Dividing the two prime-counting asymptotics for moduli Q and M gives (74). □

Corollary 6.2 (all 48 reduced 1 mod 12 classes modulo 840)**.** *For every primitive positive ray* $[r : t]$ *and every one of the* 48 *reduced classes*

$$c \pmod{840}, \qquad c \equiv 1 \pmod{12}, \tag{75}$$

there is a reduced prime progression consisting entirely, apart from a finite initial segment, of exterior certificates on that ray.

7 A single Koide ray captures density one of every progression

The preceding theorem lets R depend on the ray and progression. On the diagonal ray $[1:1]$, a stronger fixed description is available.

Let $p \equiv 1 \pmod 4$ be prime and let $q \equiv 3 \pmod 4$ be a prime divisor of $p+1$. Put

$$s = \frac{p+q}{4}, \qquad k = \frac{p+1}{q}. \tag{76}$$

Then $u = a = v = s$ and

$$\boxed{\frac{4}{p} = \frac{1}{s} + \frac{1}{sk} + \frac{1}{psk}.} \tag{77}$$

Indeed,

$$\frac{1}{s} + \frac{1}{sk} + \frac{1}{psk} = \frac{pk+p+1}{psk} = \frac{(p+1)(p+q)/q}{p(p+q)(p+1)/(4q)} = \frac{4}{p}.$$

Since $qk = p+1 \equiv 2 \pmod 4$ and $q \equiv -1 \pmod 4$, one has $k \equiv 2 \pmod 4$. This is the exterior channel with

$$R = q, \quad j = \frac{q-3}{4}, \quad T = k+1, \quad (\ell_0, \ell_1, \ell_2) = \left(\frac{k+2}{4}, \frac{3k+2}{4}, 1\right), \tag{78}$$

and terminal determinant -1.

Fix M, c as in (63). For a finite set $\mathcal{Q}$ of primes $q \equiv 3 \pmod 4$ with $q \nmid M$, let $\mathscr{P}_{\mathcal{Q}}(M,c)$ denote the primes $p \equiv c \pmod M$ for which $q \mid p+1$ for at least one $q \in \mathcal{Q}$.

Theorem 7.1 (finite sieve density and density-one diagonal ray)**.** *The relative density of $\mathscr{P}_{\mathcal{Q}}(M,c)$ inside the primes $p \equiv c \pmod M$ exists and equals*

$$\boxed{1 - \prod_{q \in \mathcal{Q}} \frac{q-2}{q-1}.} \tag{79}$$

As $\mathcal{Q}$ increases through all primes $q \equiv 3 \pmod 4$, $q \nmid M$, the relative density tends to one. Consequently the exterior certificates on the single primitive Koide ray

$$[r:t] = [1:1], \qquad \Gamma_s = s \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \tag{80}$$

capture relative density one of the primes in every reduced progression $c \pmod M$ with $c \equiv 1 \pmod 4$.

Proof. Put $Q_{\mathcal{Q}} = \prod_{q \in \mathcal{Q}} q$. Since every q is coprime to M, the Chinese remainder theorem identifies the reduced residue classes modulo $MQ_{\mathcal{Q}}$ above $c \pmod M$ with the product of the $q-1$ unit classes modulo each q. Avoiding $p \equiv -1 \pmod q$ leaves $q-2$ classes. The prime number theorem in arithmetic progressions therefore gives relative density

$$\prod_{q \in \mathcal{Q}} \frac{q-2}{q-1}$$

for the complement, proving (79).

The sum of reciprocals of primes $q \equiv 3 \pmod 4$ diverges. Hence

$$\sum_{q \equiv 3 (4)} \frac{1}{q-1} = \infty, \qquad \prod_{q \in \mathcal{Q}} \left(1 - \frac{1}{q-1}\right) \longrightarrow 0.$$

Every prime counted by the union has the explicit certificate (77); taking larger finite sets shows that the relative lower density of the union is at least every quantity in (79), hence is one. □

Corollary 7.2 (a fixed 10% subprogression modulo 9240)**.** *Take $M = 840$ and $q = 11$. In each reduced class $c \equiv 1 \pmod{12}$ modulo 840, the simultaneous congruences*

$$p \equiv c \pmod{840}, \qquad p \equiv -1 \pmod{11} \tag{81}$$

define one reduced class modulo 9240. Its primes have the diagonal-ray certificate (77) and form relative density $1/(11-1) = 1/10$ inside the class $c \pmod{840}$.

Example 7.3. For $p = 109$ and $q = 11$,

$$s = 30, \qquad k = 10,$$

so

$$\frac{4}{109} = \frac{1}{30} + \frac{1}{300} + \frac{1}{32700}. \tag{82}$$

The terminal triple is $(\ell_0, \ell_1, \ell_2) = (3, 8, 1)$ and

$$\mathsf{R} = \begin{pmatrix} 3 & -8 \\ 4 & -11 \end{pmatrix}, \qquad \det \mathsf{R} = -1.$$

Remark 7.4 (the unresolved quantifier). The theorem proves

$$\forall\, [r : t]\ \forall\, c \pmod{M} \quad \text{infinitely many primes in the class have a certificate on the ray,}$$

and, on $[1 : 1]$, a density-one statement. The Erdős–Straus conjecture asks instead

$$\forall\, p \quad \exists\, [r : t], R \quad \text{a certificate for that particular prime.}$$

A density-zero exceptional set is not removed by theorem 7.1.

8 The universal diagonal divisor identity

Retain the complete coprime odd shell (16) and restrict to the primitive ray $[r : t] = [1 : 1]$. Then $u = a = v = s$ with $s = (n + R)/4$.

Proposition 8.1 (all diagonal divisor channels)**.** *For $\delta \mid n^2$, the diagonal ray supports the oriented channel of order δ exactly when*

$$R \mid n + \delta. \tag{83}$$

Put

$$k_\delta = \frac{n + \delta}{R}. \tag{84}$$

Then the corresponding reciprocal decomposition is

$$\boxed{\frac{4}{n} = \frac{1}{s} + \frac{1}{sk_\delta} + \frac{1}{snk_\delta/\delta}, \qquad s = \frac{n+R}{4}.} \tag{85}$$

The third displayed denominator is an integer. The terminal coordinates are

$$T_\delta = k_\delta + 1, \qquad \ell_2 = \sigma_\delta, \qquad \ell_0 = \frac{k_\delta + 1 + \sigma_\delta}{4}, \qquad \ell_1 = \frac{3k_\delta + 3 - \delta}{4}, \tag{86}$$

and their terminal determinant is $-\delta$.

Under the complementary orientation $\delta' = n^2/\delta$, one has

$$k_{\delta'} = \frac{n}{\delta} k_\delta, \tag{87}$$

and the final two denominators in (85) are exchanged exactly.

Proof. On $[1 : 1]$, equation (51) is $R \mid n + \delta$, proving the first assertion. The channel reconstruction (24) gives

$$y = \frac{ns + (n^2/\delta)s}{R} = \frac{snk_\delta}{\delta}, \qquad z = \frac{ns + \delta s}{R} = sk_\delta.$$

Their integrality is already part of the complete channel theorem; direct addition gives

$$\frac{1}{s} + \frac{1}{sk_\delta} + \frac{\delta}{snk_\delta} = \frac{nk_\delta + n + \delta}{snk_\delta} = \frac{k_\delta(n + R)}{snk_\delta} = \frac{4}{n}.$$

Because $T_\delta = (4s + \delta)/R$, one has $T_\delta = k_\delta + 1$. Substitution in (36) gives (86); its integrality and determinant are those of theorem 3.1. Finally,

$$k_{n^2/\delta} = \frac{n + n^2/\delta}{R} = \frac{n}{\delta}\frac{n + \delta}{R},$$

and substitution swaps sk_δ with snk_δ/δ. □

9 Exact multiplicity on the diagonal Koide ray

Let $p \equiv 1 \pmod 4$ be prime and put

$$m_p = \frac{p + 1}{2}. \tag{88}$$

Thus m_p is odd. Let χ_4 denote the nontrivial real Dirichlet character modulo four on odd integers.

Theorem 9.1 (complete diagonal-ray certificate count)**.** *The map*

$$R \longmapsto (x_R, y_R, z_R) = \left(\frac{p + R}{4}, \frac{p + R}{4}\frac{p + 1}{R}, p\frac{p + R}{4}\frac{p + 1}{R}\right) \tag{89}$$

is a bijection from

$$\mathscr{R}_{\mathrm{diag}}(p) = \{R > 0 : R \mid m_p,\ R \equiv 3 \pmod 4\} \tag{90}$$

onto the exterior Erdős–Straus selector states on the exact primitive ray $[r : t] = [1 : 1]$. Every displayed triple is ordered and satisfies

$$\frac{4}{p} = \frac{1}{x_R} + \frac{1}{y_R} + \frac{1}{z_R}. \tag{91}$$

The number of these states is

$$\boxed{\nu_{\mathrm{diag}}(p) = \frac{1}{2}\left(\tau(m_p) - \sum_{d \mid m_p} \chi_4(d)\right).} \tag{92}$$

If

$$m_p = \prod_{q\equiv 1(4)} q^{e_q} \prod_{r\equiv 3(4)} r^{f_r}, \tag{93}$$

then

$$\boxed{\nu_{\mathrm{diag}}(p) = \frac{1}{2}\left[\prod_{q\equiv 1(4)} (e_q+1) \prod_{r\equiv 3(4)} (f_r+1) - \prod_{q\equiv 1(4)} (e_q+1) \prod_{r\equiv 3(4)} \epsilon(f_r)\right],} \tag{94}$$

where $\epsilon(f) = 1$ *for even* f *and* $\epsilon(f) = 0$ *for odd* f*. Consequently*

$$\boxed{\nu_{\mathrm{diag}}(p) = 0 \quad\Longleftrightarrow\quad \textit{every prime divisor of } (p+1)/2 \textit{ is } 1 \pmod 4.} \tag{95}$$

Proof. On the ray $[1:1]$, the canonical factor coordinates are $u = a = v = s$. The exterior projective incidence (51) becomes $R \mid p+1$, while the shell equation $4a = p + R$ gives $s = (p+R)/4$. Because R is odd, $R \mid p+1$ is equivalent to $R \mid m_p$. Conversely, every divisor in (90) satisfies $3 \le R \le m_p \le p-2$ and defines one allowed shell marker. With $k_R = (p+1)/R$, the source exterior reconstruction is exactly $(x_R, y_R, z_R) = (s, sk_R, psk_R)$, proving the bijection and (91). Since $k_R \ge 2$, the triple is ordered. Distinct values of R give distinct first denominators.

For every odd divisor d of m_p, the indicator of $d \equiv 3 \pmod 4$ is $(1 - \chi_4(d))/2$. Summation gives (92). Multiplicativity gives

$$\sum_{d\mid m_p} \chi_4(d) = \prod_{q\equiv 1(4)} (e_q+1) \prod_{r\equiv 3(4)} \sum_{j=0}^{f_r} (-1)^j,$$

and the last geometric sum is $\epsilon(f_r)$. This proves (94). The right side vanishes exactly when the second prime product in (93) is empty, proving (95). □

Corollary 9.2 (exact terminal multiplicity)**.** *For* $R \in \mathscr{R}_{\mathrm{diag}}(p)$ *put*

$$k_R = \frac{p+1}{R} \equiv 2 \pmod 4. \tag{96}$$

The corresponding terminal coordinates and matrix are

$$(\ell_0, \ell_1, \ell_2) = \left(\frac{k_R+2}{4}, \frac{3k_R+2}{4}, 1\right), \qquad \mathsf{R}_R = \begin{pmatrix} 3 & -(3k_R+2)/4 \\ 4 & -(k_R+1) \end{pmatrix}, \tag{97}$$

with $\det \mathsf{R}_R = -1$*. These matrices are pairwise distinct as* R *varies. Hence* $\nu_{\mathrm{diag}}(p)$ *is simultaneously the number of ordered diagonal-ray decompositions, diagonal-ray exterior selector states, and determinant-*(-1) *terminal matrices attached to this ray.*

Proof. The formulas are (78) with q replaced by the full allowed divisor R. Distinct R give distinct $k_R = (p+1)/R$, and the lower-right matrix entry is $-(k_R+1)$. □

Corollary 9.3 (the exact hard residue split)**.** *Every prime* $p \equiv 5 \pmod 8$ *has at least one diagonal-ray certificate. In particular every hard prime* $p \equiv 13 \pmod{24}$ *is captured by this one ray. A hard prime not captured by the ray must satisfy* $p \equiv 1 \pmod{24}$ *and* $(p+1)/2$ *must lie in the multiplicative semigroup generated by primes congruent to* 1 *modulo four.*

Proof. If $p \equiv 5 \pmod 8$, then $m_p \equiv 3 \pmod 4$, so its prime factorization contains a prime congruent to 3 modulo four. Apply (95). The final assertion follows by combining that equivalence with the two possible hard classes modulo 24. □

Example 9.4 (the two diagonal certificates at $p = 109$)**.** Here $m_p = 55 = 5 \cdot 11$, and its divisors congruent to 3 modulo four are 11 and 55. Formula (89) gives the two distinct ordered solutions

$$\frac{4}{109} = \frac{1}{30} + \frac{1}{300} + \frac{1}{32700} = \frac{1}{41} + \frac{1}{82} + \frac{1}{8938}.$$

Their terminal matrices are respectively

$$\begin{pmatrix} 3 & -8 \\ 4 & -11 \end{pmatrix}, \qquad \begin{pmatrix} 3 & -2 \\ 4 & -3 \end{pmatrix},$$

and both have determinant -1.

10 Middle-channel rays

For $n = p$ prime and $\delta = p$, (51) becomes

$$R \mid r + t. \tag{98}$$

Thus a primitive ray admits a middle channel at precisely the divisors

$$\mathscr{R}_M(r, t) = \{R \mid r + t : R \equiv 3 \pmod 4\}. \tag{99}$$

This set is nonempty exactly when $r + t$ has a prime divisor congruent to 3 (mod 4).

For fixed $R \in \mathscr{R}_M(r, t)$, put $A = 4rt$, $g = \gcd(A, M)$ and $M' = M/g$.

Theorem 10.1 (middle-ray prime progressions)**.** *A prescribed prime class $p \equiv c \pmod M$ is compatible with this ray and R exactly when*

$$g \mid c + R. \tag{100}$$

Under this condition the congruence

$$\frac{A}{g} s_0 \equiv \frac{c + R}{g} \pmod{M'} \tag{101}$$

has one class modulo M'. Put

$$p_0 = As_0 - R, \qquad Q_M = AM'. \tag{102}$$

Then $\gcd(p_0, Q_M) = 1$, and every sufficiently large prime $p \equiv p_0 \pmod{Q_M}$ lies in the class $c \pmod M$ and gives

$$\boxed{x = rst, \qquad y = prs\,\frac{r + t}{R}, \qquad z = pst\,\frac{r + t}{R},} \tag{103}$$

where $s = (p + R)/(4rt)$. These primes have asymptotic count $\mathrm{Li}(X)/\varphi(Q_M)$.

Proof. The linear congruence $As \equiv c + R \pmod M$ is solvable exactly under (100); division by g gives (101). Since $R \mid r + t$ and $\gcd(r, t) = 1$, one has $\gcd(R, rt) = 1$, so $\gcd(R, A) = 1$. The residue p_0 is a unit modulo A because $p_0 \equiv -R \pmod A$, and it is a unit modulo M' because $p_0 \equiv c \pmod M$. Therefore $\gcd(p_0, Q_M) = 1$, and the prime number theorem in arithmetic progressions applies. For every such prime, $p = 4rts - R$ and $R \mid r + t$ give the middle incidence. Substitution into the source factor-coordinate reconstruction gives (103). □

Unlike the exterior theorem, the allowed residuals are the finite set (99) attached to the ray.

11 Zariski density on the determinant cone

Let

$$\mathscr{C}_{\det} = \{(u, a, v) \in \mathbb{A}^3 : uv - a^2 = 0\}. \tag{104}$$

Its positive rational projective rays are

$$[r : t] \longmapsto [r^2 : rt : t^2]. \tag{105}$$

Theorem 11.1 (Zariski density of exterior certificate matrices)**.** *Fix M, c as in (63). The matrices*

$$\Gamma = s \begin{pmatrix} r^2 & rt \\ rt & t^2 \end{pmatrix} \tag{106}$$

arising from the exterior prime progressions of theorem 6.1, as $[r : t]$ ranges over primitive positive rays, are Zariski dense in $\mathscr{C}_{\det}$. Their images are Zariski dense in the corresponding native and raw determinant-zero coefficient cones under the invertible linear maps (11) and (13).

Proof. Let $F(u, a, v)$ vanish on every constructed matrix. Fix a primitive ray (r, t). The prime-saturation theorem produces infinitely many distinct positive contents s on that ray, so

$$F(sr^2, srt, st^2) = 0$$

for infinitely many s. This is a polynomial in s and therefore vanishes identically. Hence every homogeneous component of F vanishes at $[r^2 : rt : t^2]$. The positive rational values of t/r are Zariski dense in $\mathbb{P}^1$, and the Veronese image (105) is the irreducible projective conic underlying (104). Thus every homogeneous component vanishes on the cone, and so does F. Invertible linear maps preserve Zariski density. □

12 Bézout openings of null rays

The selector matrix (53) is singular. The next construction adds an exact integral transverse direction instead of declaring the null ray to be a regular period point.

Fix a primitive positive vector

$$\mathbf{v} = \begin{pmatrix} r \\ t \end{pmatrix}. \tag{107}$$

A Bézout companion is a vector

$$\mathbf{w} = \begin{pmatrix} x \\ y \end{pmatrix}, \qquad ry - tx = 1. \tag{108}$$

Then

$$B_{\mathbf{v},\mathbf{w}} = \begin{pmatrix} r & x \\ t & y \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z}). \tag{109}$$

Definition 12.1 (Bézout opening)**.** For $\delta > 0$, define

$$\mathsf{A}_{\delta,\mathbf{w}} = \delta \mathbf{v}\mathbf{v}^{\mathsf{T}} - \mathbf{w}\mathbf{w}^{\mathsf{T}} = B_{\mathbf{v},\mathbf{w}} \operatorname{diag}(\delta, -1) B_{\mathbf{v},\mathbf{w}}^{\mathsf{T}}. \tag{110}$$

Theorem 12.2 (integral opening and regular ambient period point)**.** *Every opening satisfies*

$$\det \mathsf{A}_{\delta,\mathbf{w}} = -\delta. \tag{111}$$

The companions are exactly $\mathbf{w}_k = \mathbf{w} + k\mathbf{v}$*,* $k \in \mathbb{Z}$*, and*

$$\mathsf{A}_{\delta,\mathbf{w}_k} = \mathsf{A}_{\delta,\mathbf{w}} - k(\mathbf{v}\mathbf{w}^{\mathsf{T}} + \mathbf{w}\mathbf{v}^{\mathsf{T}}) - k^2\mathbf{v}\mathbf{v}^{\mathsf{T}}. \tag{112}$$

The map $k \mapsto \mathsf{A}_{\delta,\mathbf{w}_k}$ *is injective.*

One may choose the companion so that $|x| \le r/2$*. For that choice,*

$$(\mathsf{A}_{\delta,\mathbf{w}})_{11} = \delta r^2 - x^2 \ge \left(\delta - \frac{1}{4}\right) r^2 > 0. \tag{113}$$

Write

$$\mathsf{A}_{\delta,\mathbf{w}} = \begin{pmatrix} A_{11} & A_{12} \\ A_{12} & A_{22} \end{pmatrix}. \tag{114}$$

Then the raw coefficient triple

$$(T_{\mathrm{raw}}, M_{\mathrm{raw}}, B_{\mathrm{raw}}) = \left(A_{11}, \frac{A_{12}}{\sqrt{6}}, A_{22}\right) \tag{115}$$

has

$$T_{\mathrm{raw}} B_{\mathrm{raw}} - 6M_{\mathrm{raw}}^2 = -\delta, \qquad T_{\mathrm{raw}} > 0, \tag{116}$$

and

$$B_{\mathrm{raw}} - \frac{6M_{\mathrm{raw}}^2}{T_{\mathrm{raw}}} = -\frac{\delta}{A_{11}} < 0. \tag{117}$$

Thus the purely imaginary periods

$$\tau = iA_{11}, \qquad \mu = i\frac{A_{12}}{\sqrt{6}}, \qquad \beta = iA_{22} \tag{118}$$

give

$$Z_A = i\begin{pmatrix} \sqrt{6}A_{12} & A_{11} \\ A_{22} & A_{12}/\sqrt{6} \end{pmatrix}, \qquad P_A = \begin{pmatrix} 0 & 0 & 1 & 0 \\ \sqrt{6}A_{12} & A_{11} & 0 & 0 \\ 0 & 0 & 0 & 1 \\ A_{22} & A_{12}/\sqrt{6} & 0 & 0 \end{pmatrix}, \tag{119}$$

with

$$\det P_A = A_{11}A_{22} - A_{12}^2 = -\delta. \tag{120}$$

Thus they give a full-rank lattice in the canonical ambient period form. The native coefficient triple is

$$(t, m, b) = \left(\frac{A_{11}}{2}, \frac{A_{12}}{2}, 3A_{22}\right), \tag{121}$$

for which

$$tb - 6m^2 = -\frac{3}{2}\delta, \qquad G(t, m, b) = \mathsf{A}_{\delta,\mathbf{w}}. \tag{122}$$

The two triples are related exactly by (15)*.*

Proof. Equation (111) follows from $\det B_{\mathbf{v},\mathbf{w}} = 1$. Every other solution of $ry - tx = 1$ differs by an integral multiple of (r, t), and direct expansion gives (112). In the basis $(\mathbf{v}, \mathbf{w})$, the opening has matrix

$$\begin{pmatrix} \delta - k^2 & -k \\ -k & -1 \end{pmatrix},$$

so equality for two integers k forces equality of their off-diagonal entries. Shifting the companion chooses a centered representative of x modulo r, proving (113).

Equations (116) and (117) are the determinant identity divided by the positive entry A_{11}. In the canonical realification of $Z = \left(\begin{smallmatrix} 6\mu & \tau \\ \beta & \mu \end{smallmatrix}\right)$, the real period determinant is exactly the displayed determinant (120), equivalently $T_{\mathrm{raw}} B_{\mathrm{raw}} - 6M_{\mathrm{raw}}^2 = -\delta \neq 0$. Therefore its four columns span $\mathbb{R}^4$ and form a lattice. Substitution into (10) gives (122); applying (15) gives (115). □

Remark 12.3 (exact geometric scope). The opening constructs a regular complex two-torus in the ambient canonical period domain. It supplies neither the real parts nor the holomorphic and triangle-group equivariance conditions needed to place that point on the global period image of the proposed S^6 family. The obstruction is not rank: the rank defect has been removed exactly. The remaining question is membership in that particular global period map.

Corollary 12.4 (opening is constant along a saturated prime ray)**.** *Fix (r, t), δ, and a Bézout companion. The matrix $\mathsf{A}_{\delta,\mathbf{w}}$ is independent of the content s, the external denominator n, and the residual R. Thus every infinite prime family produced on the fixed ray by theorem 6.1 maps to one fixed regular ambient period point after the opening data are fixed.*

13 Exact left–right morphism to the terminal matrix

Let R_δ be the terminal matrix in (38). Define

$$P_0 = \begin{pmatrix} -1 & 1 \\ -4 & 3 \end{pmatrix}, \qquad Q_0 = \begin{pmatrix} 1 & T_\delta - \ell_1 \\ 0 & 1 \end{pmatrix}. \tag{123}$$

Then

$$P_0 \mathsf{R}_\delta Q_0 = \mathrm{diag}(1, -\delta). \tag{124}$$

For $B = B_{\mathbf{v},\mathbf{w}}$ and $J = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, put

$$L_A = -JB^{-1}, \qquad R_A = B^{-\mathsf{T}} J. \tag{125}$$

Both are in $\mathrm{GL}_2(\mathbb{Z})$ and

$$L_A \mathsf{A}_{\delta,\mathbf{w}} R_A = \mathrm{diag}(1, -\delta). \tag{126}$$

Theorem 13.1 (coordinate-level opening–terminal morphism)**.** *The terminal and opened matrices are related by the exact left–right identity*

$$\boxed{\mathsf{R}_\delta = P_0^{-1} L_A \, \mathsf{A}_{\delta,\mathbf{w}} \, R_A Q_0^{-1}.} \tag{127}$$

Every displayed left and right factor belongs to $\mathrm{GL}_2(\mathbb{Z})$ and has an explicit inverse. Thus the relationship is stronger than equality of determinants or Smith invariants.

Proof. A direct multiplication gives

$$P_0 \begin{pmatrix} 3 & -\ell_1 \\ 4 & -T_\delta \end{pmatrix} = \begin{pmatrix} 1 & \ell_1 - T_\delta \\ 0 & 4\ell_1 - 3T_\delta \end{pmatrix} = \begin{pmatrix} 1 & \ell_1 - T_\delta \\ 0 & -\delta \end{pmatrix}.$$

Right multiplication by Q_0 gives (124). Equation (110) gives

$$\begin{aligned} L_A \mathsf{A}_{\delta,\mathbf{w}} R_A &= -JB^{-1} B \, \mathrm{diag}(\delta, -1) B^{\mathsf{T}} B^{-\mathsf{T}} J \\ &= -J \, \mathrm{diag}(\delta, -1) J = \mathrm{diag}(1, -\delta). \end{aligned}$$

Equating the two diagonalizations proves (127). □

Example 13.2 (the fixed diagonal-ray opening)**.** For $\mathbf{v} = (1,1)^{\mathsf{T}}$ choose $\mathbf{w} = (0,1)^{\mathsf{T}}$. At $\delta = 1$,

$$\mathsf{A}_* = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \det \mathsf{A}_* = -1. \tag{128}$$

The raw period triple is

$$(T_{\mathrm{raw}}, M_{\mathrm{raw}}, B_{\mathrm{raw}}) = \left(1, \frac{1}{\sqrt{6}}, 0\right).$$

For the density-one family (78), every varying terminal matrix is $\mathrm{GL}_2(\mathbb{Z})$-equivalent through (127) to this one fixed ambient opening.

14 Exact map to the defect-one Koide shell

For an opening $A = \mathsf{A}_{\delta,\mathbf{w}}$, define its Lorentz coordinate vector

$$\mathbf{z}(A) = \left(\frac{A_{11} + A_{22}}{2}, A_{12}, \frac{A_{11} - A_{22}}{2}\right) = (H_A, X_A, Y_A). \tag{129}$$

By construction,

$$H_A^2 - X_A^2 - Y_A^2 = \det A = -\delta. \tag{130}$$

Define the exact homothety and orthogonal inverse

$$\mathfrak{K}_\delta(A) = \mathcal{F}^{-1}\left(\frac{\mathbf{z}(A)}{\sqrt{3\delta}}\right). \tag{131}$$

Theorem 14.1 (channel opening to the defect-one Koide hyperboloid)**.** *For every divisor channel order $\delta > 0$ and every Bézout opening,*

$$Q_K(\mathfrak{K}_\delta(A)) = -\frac{1}{3}. \tag{132}$$

Equivalently, with

$$\Delta(q) = 4(q_1q_2 + q_1q_3 + q_2q_3) - (q_1^2 + q_2^2 + q_3^2) = 3Q_K(q), \tag{133}$$

one has

$$\boxed{\Delta(\mathfrak{K}_\delta(A)) = -1.} \tag{134}$$

The map is defined over $\mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{\delta})$ for integral A. It need not land in the integral defect-one shell.

Proof. The map $\mathcal{F}$ is orthogonal and identifies Q_K with the Lorentz form. Equations (130) and (131) therefore give

$$Q_K(\mathfrak{K}_\delta(A)) = \frac{-\delta}{3\delta} = -\frac{1}{3}.$$

The identity $\Delta = 3Q_K$ gives (134). The displayed radicals in $\mathcal{F}^{-1}$ and in the homothety give the stated field. □

Theorem 14.2 (exact rational nonintersection of the orthogonal defect map)**.** *Let $\delta \in \mathbb{Q}_{>0}$ and let $A \in \mathrm{Sym}_2(\mathbb{Q})$ satisfy $\det A = -\delta$. Then*

$$\boxed{\mathfrak{K}_\delta(A) \notin \mathbb{Q}^3.} \tag{135}$$

Consequently the image of the integral Bézout openings under the exact map (131) is disjoint from the integral defect-one shell, not merely unproved to lie in it.

Proof. Write $\mathbf{z}(A) = (H_A, X_A, Y_A)$ and suppose $q = \mathfrak{K}_\delta(A) = (q_1, q_2, q_3)$ were rational. Put

$$S = q_1 + q_2 + q_3, \qquad U = q_1 - q_3, \qquad V = q_1 + q_3 - 2q_2. \tag{136}$$

Applying the exact orthogonal map (9) to $q = \mathcal{F}^{-1}(\mathbf{z}(A)/\sqrt{3\delta})$ gives

$$H_A = \sqrt{\delta}\, S, \qquad X_A = \sqrt{\frac{3\delta}{2}}\, U, \qquad Y_A = \sqrt{\frac{\delta}{2}}\, V. \tag{137}$$

All six quantities H_A, X_A, Y_A, S, U, V are rational.

If two among H_A, X_A, Y_A are nonzero, division of the corresponding two relations in (137) would make one of $\sqrt{3/2}$, $1/\sqrt{2}$, or $\sqrt{3}$ rational. Hence at most one of H_A, X_A, Y_A is nonzero. The case in which only H_A is nonzero is impossible because $H_A^2 - X_A^2 - Y_A^2 = -\delta < 0$. If only X_A is nonzero, then $X_A^2 = \delta$ and the second relation gives $U^2 = 2/3$, impossible for $U \in \mathbb{Q}$. If only Y_A is nonzero, then $Y_A^2 = \delta$ and the third relation gives $V^2 = 2$, again impossible over $\mathbb{Q}$. The zero triple contradicts $\det A = -\delta$. Thus no rational q exists. □

Remark 14.3 (exact separation from the counted integral orbit). The map (131) proves equality of the quadratic level $\Delta = -1$ and produces a real algebraic point for every opening. Theorem 14.2 proves that this particular map has empty intersection with $\mathbb{Q}^3$, hence with the integral chamber counted in the Koide defect-one theorem. Any arithmetic transfer to that counted orbit requires a different, explicitly constructed morphism.

Remark 14.4 (why the existing count does not transfer). The current Koide arithmetic workbench counts integral points on a particular reduced chamber of $\Delta = -1$ and proves a linear asymptotic there. The map (131) produces exact algebraic points and, by theorem 14.2, none of them is rational. Therefore the proved integral counting constant is not a count of Erdős–Straus selector states. What is transferred is the quadratic level and the complete coordinate map, not integrality or chamber membership.

15 Faithful certificates and exact inverse data

An oriented divisor channel determines the pair

$$\mathfrak{C}(n, R, \delta, u, v) = \left(\Gamma = \begin{pmatrix} u & a \\ a & v \end{pmatrix}, \quad \mathsf{R}_\delta = \begin{pmatrix} 3 & -\ell_1 \\ 4 & -T_\delta \end{pmatrix} \right). \tag{138}$$

The two determinant equations are

$$\det \Gamma = 0, \qquad \det \mathsf{R}_\delta = -\delta, \tag{139}$$

and the coupling equations are

$$4a = n + R, \qquad RT_\delta = 4u + \delta, \qquad uv = a^2. \tag{140}$$

Theorem 15.1 (faithfulness of the double certificate)**.** *The map from marked oriented channel states to the pair* (138) *is injective. The inverse is*

$$u = \Gamma_{11}, \qquad a = \Gamma_{12}, \qquad v = \Gamma_{22}, \tag{141}$$

$$T_\delta = -(\mathsf{R}_\delta)_{22}, \qquad \ell_1 = -(\mathsf{R}_\delta)_{12}, \tag{142}$$

$$j = \frac{u - \ell_1}{T_\delta}, \qquad R = 4j + 3, \tag{143}$$

$$n = 4a - R, \qquad \delta = -\det \mathsf{R}_\delta. \tag{144}$$

The terminal matrix alone is not faithful; the rank-one matrix restores the external denominator, shell marker, divisor support, and channel orientation.

Proof. Equations (141) and (142) read off matrix entries. The definition $\ell_1 = u - jT_\delta$ gives (143); then (17) and (39) give (144). Substitution reconstructs every original coordinate. □

16 Exact consequences and remaining boundary

The proved chain is

$$\boxed{\begin{array}{c}\text{oriented fixed-shell reciprocal state} \longleftrightarrow (\delta, u, v),\\ (\delta, u, v) \longmapsto (\Gamma, \mathsf{R}_\delta)\,,\\ \Gamma \longmapsto [\mathbf{v}], \qquad ([\mathbf{v}], \delta, \mathbf{w}) \longmapsto \mathsf{A}_{\delta,\mathbf{w}},\\ \mathsf{A}_{\delta,\mathbf{w}} \xrightarrow{\mathrm{GL}_2(\mathbb{Z})\times\mathrm{GL}_2(\mathbb{Z})} \mathsf{R}_\delta,\\ \mathsf{A}_{\delta,\mathbf{w}} \longmapsto \text{regular ambient period point},\\ \mathsf{A}_{\delta,\mathbf{w}} \xrightarrow{\mathfrak{K}_\delta} \{\Delta = -1\}.\end{array}} \tag{145}$$

Every arrow has been displayed in coordinates and supplied with its inverse when it is bijective.

The following statements are exact consequences.

(i) The prime two-channel selector is the specialization of a complete $\delta \mid n^2$ channel theory.

(ii) Complementary denominator orientations correspond to reciprocal terminal orders whose product is n^2, and to opposite Lorentz rapidities.

(iii) Every primitive positive determinant-null ray supports infinitely many exterior prime certificates in every admissible prime progression.

(iv) One ray, $[1:1]$, supports relative density one of those primes; its certificate multiplicity for each individual prime is exactly the divisor count in (92), and it carries a fixed determinant-(-1) ambient opening.

(v) Every channel ray has infinitely many Bézout openings of determinant $-\delta$, and each opening is connected to its terminal matrix by the exact left–right identity (127).

(vi) Every opening maps to the real defect-one Koide hyperboloid, but the exact orthogonal–homothetic image is disjoint from $\mathbb{Q}^3$; no integral point count is transferred by this morphism.

The Erdős–Straus conjecture remains the surjectivity statement in the other quantifier direction: every prime must lie in at least one selector fibre. Neither density one, Zariski density, nor infinite saturation of every fixed ray rules out a density-zero set of primes missed by all rays.

Computational receipt

The accompanying exact-arithmetic verifier checks:

- the channel bijection and fixed-point-free complement in 665 coprime shells with $n \le 161$;
- 1430 oriented channel states and 512 explicit opening–terminal $\mathrm{GL}_2(\mathbb{Z})$ morphisms;
- the prime specialization in 9137 shells for primes $p < 1000$;

- 57 character-sum channel counts;
- 192 primitive-ray/congruence-class progressions and explicit prime certificates in all 48 reduced classes $c \equiv 1 \pmod{12}$ modulo 840 on the diagonal ray;
- 30 Bézout-opening torsors with eleven consecutive companion shifts each;
- the complete diagonal-ray divisor count for every prime $p < 5000$ with $p \equiv 1 \pmod{4}$, including the exact terminal matrix multiplicity.

These checks are bounded receipts for the displayed identities. The proofs above establish the unbounded statements.

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